

JanusCoder - Code Intelligence with Vision - Run Locally



Fahd Mirza
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This video locally installs JanusCoderV, a model designed to establish a unified visual-programmatic interface for code intelligence.

https://huggingface.co/internlm/JanusCoder-8B

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Introduction

We introduce JanusCoder and JanusCoderV, a suite of open-source foundational models designed to establish a unified visual-programmatic interface for code intelligence. This model suite is built upon open-source language models (such as Qwen3-8B and

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JanusCoderV-8B

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Introduction

We introduce JanusCoder and JanusCoderV, a suite of open-source foundational models designed to establish a unified visual-programmatic interface for code intelligence. This model suite is built upon open-source language models (such as Qwen3-8B and 14B) and multimodal models (such as Qwen2.5-VL and

Downloads last month

Safetensors Model size 9B params Tensor type BF16 Chat template Files info

Inference Providers NEW Image-Text-to-Text This model isn't deployed by any Inference Provider. Ask for provider support

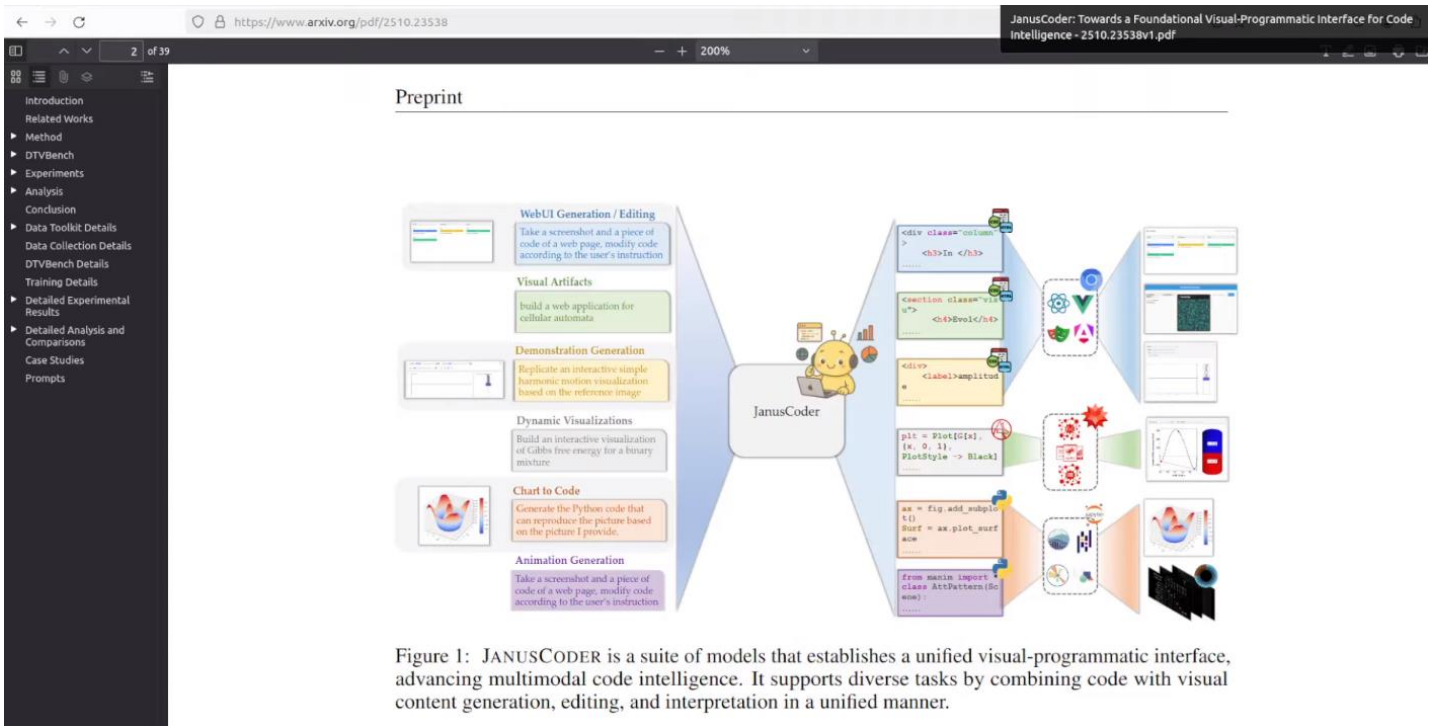
```
+-----+
ubuntu@0054-dsm2-sma6-prxm70017:~$ conda create -n ai python=3.11 -y && conda activate ai
```

```
(ai) ubuntu@0054-dsm2-sma6-prxmx70017:~$ pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
pip install einops timm pillow
pip install git+https://github.com/huggingface/transformers
Looking in indexes: https://download.pytorch.org/whl/cu121
Collecting torch
  Using cached https://download.pytorch.org/whl/cu121/torch-2.5.1%2Bcu121-cp311-cp311-linux_x86_64.whl (780.5 MB)
```

```
ubuntu@0054-dsm2-sma6-prxmx70017:~$ hf auth login
User is already logged in.
(ai) ubuntu@0054-dsm2-sma6-prxmx70017:~$
```

```
ubuntu@0054-dsm2-sma6-prxmx70017:~$ hf download internlm/JanusCoderV-8B
Fetching 19 files: 0% | 0/19 [00:00<?, ?it/s]
Downloading 'generation_config.json' to '/home/ubuntu/.cache/huggingface/hub/models--internlm--JanusCoderV-8B/blobs/4f0c4ce3e3028b76812135c422c721c3e77b10b6.incomplete'
```

```
Download complete. Moving file to /home/ubuntu/.cache/huggingface/hub/models--internlm--JanusCoderV-8B/blobs/4783fe10ac3adce15ac8f358ef5462739852c569 | 30.2k/11.4M [00:00<02:20, 81.1kB/s]
vocab.json: 0.00B [00:00, ?B/s]
model-00001-of-00004.safetensors: 18% | 873M/4.91G [00:05<00:15, 265MB/s]
model-00004-of-00004.safetensors: 3% | 67.0M/2.32G [00:05<03:05, 12.1MB/s]
model-00003-of-00004.safetensors: 1% | 67.6M/4.92G [00:05<06:27, 12.5MB/s]
model-00002-of-00004.safetensors: 3% | 135M/4.92G [00:05<02:49, 28.1MB/s]
```



```
Fetching 19 files: 100% | 19/19 [00:57<00:00, 3.01s/it]
/home/ubuntu/.cache/huggingface/hub/models--internlm--JanusCoderV-8B/snapshots/eadc793f2f6ce6b3299c7e4c928926755d6cade3
(ai) ubuntu@0054-dsm2-sma6-prxmx70017:~$ conda install -c conda-forge --override-channels notebook -y
conda install -c conda-forge --override-channels ipywidgets -y
jupyter notebook
```

```
localhost:8888/notebooks/Untitled6.ipynb 150%
Jupyter Untitled6 Last Checkpoint: 8 minutes ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

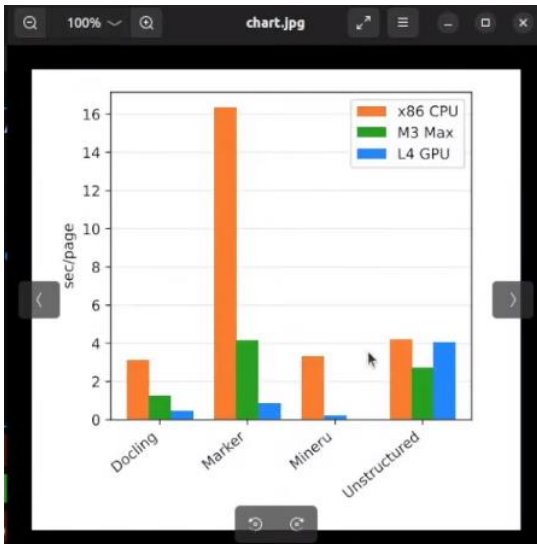
In [2]: from transformers import AutoProcessor, AutoModel
import torch
from PIL import Image

model_name = "internlm/JanusCoderV-8B"
processor = AutoProcessor.from_pretrained(model_name, trust_remote_code=True)
model = AutoModel.from_pretrained(
    model_name,
    trust_remote_code=True,
    torch_dtype=torch.bfloat16,
    device_map="auto",
).eval()

`torch_dtype` is deprecated! Use `dtype` instead!
Loading checkpoint shards: 100% 4/4 [00:01<00:00, 2.36it/s]
Some parameters are on the meta device because they were offloaded to the cpu.

Out [ ]:
```

The model is now loaded successfully



```
localhost:8888/notebooks/Untitled6.ipynb 150%
Jupyter Untitled6 Last Checkpoint: 9 minutes ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

In [ ]: image_path = "/home/Ubuntu/TransferData/images/ocrhard/chart.jpg"
image = load_image(image_path)
prompt="""
Generate the python code, which can reproduce the picture, which I have provided in image.
"""

messages = [{"role": "user", "content": [{"type": "image", "image": image}, {"type": "text", "text": prompt}],}]

inputs = processor.apply_chat_template(messages, add_generation_prompt=True, tokenize=True, return_dict=True, return_tensors="pt").to(model.device)

generate_ids = model.generate(**inputs, max_new_tokens=32768)
decoded_output = processor.decode(generate_ids[0, inputs["input_ids"].shape[1]:], skip_special_tokens=True)
print(decoded_output)
```



```
jupyter Untitled6 Last Checkpoint: 18 minutes ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

messages = [{"role": "user", "content": [{"type": "image", "image": image}, {"type": "text", "text": prompt}],}]
inputs = processor.apply_chat_template(messages, add_generation_prompt=True, tokenize=True, return_dict=True, return_tensors="pt").to(model.device)
generate_ids = model.generate(**inputs, max_new_tokens=32768)
decoded_output = processor.decode(generate_ids[0, inputs["input_ids"].shape[1]:], skip_special_tokens=True)
print(decoded_output)

x = np.arange(len(cats))
w = 0.23

plt.figure(figsize=(5, 4.8), dpi=120)
plt.bar(x - w, x86, width=w, label="x86 CPU", color="#F28E2B")
plt.bar(x, m3, width=w, label="M3 Max", color="#59A14F")
plt.bar(x + w, l4, width=w, label="L4 GPU", color="#4E79A7")

plt.ylabel("sec/page")
plt.xticks(x, cats, rotation=20)
plt.ylim(0, 17)
plt.legend(loc="upper right", frameon=True)

plt.grid(axis="y", linestyle="-.", linewidth=0.6, alpha=0.4)
for spine in ["top", "right"]:
    plt.gca().spines[spine].set_visible(False)

plt.tight_layout()
plt.show()
```

Let us now verify by running the generated code

```
[ ]: !pip install matplotlib numpy

Requirement already satisfied: six>=1.5 in /anaconda5/envs/ai/tib/python3.11/site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.10.7-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7 MB)
8.7/8.7 MB 74.7 MB/s 0:00:00
Downloading contourpy-1.3.3-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (355 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.60.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.0 MB)
5.0/5.0 MB 75.5 MB/s 0:00:00
Downloading kiwisolver-1.4.9-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.4 MB)
1.4/1.4 MB 67.0 MB/s 0:00:00
Using cached pyparsing-3.2.5-py3-none-any.whl (113 kB)
Installing collected packages: pyparsing, kiwisolver, fonttools, cycler, contourpy, matplotlib
6/6 [matplotlib] 5/6 [matplotlib]
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.60.1 kiwisolver-1.4.9 matplotlib-3.10.7 pyparsing-3.2.5

[ ]: import matplotlib.pyplot as plt
import numpy as np

cats = ["Docling", "Marker", "Mineru", "Unstructured"]
x86 = [3.1, 16.2, 3.3, 4.2]
m3 = [1.2, 4.2, 0.1, 2.8]
l4 = [0.5, 0.9, 0.1, 4.0]

x = np.arange(len(cats))
w = 0.23

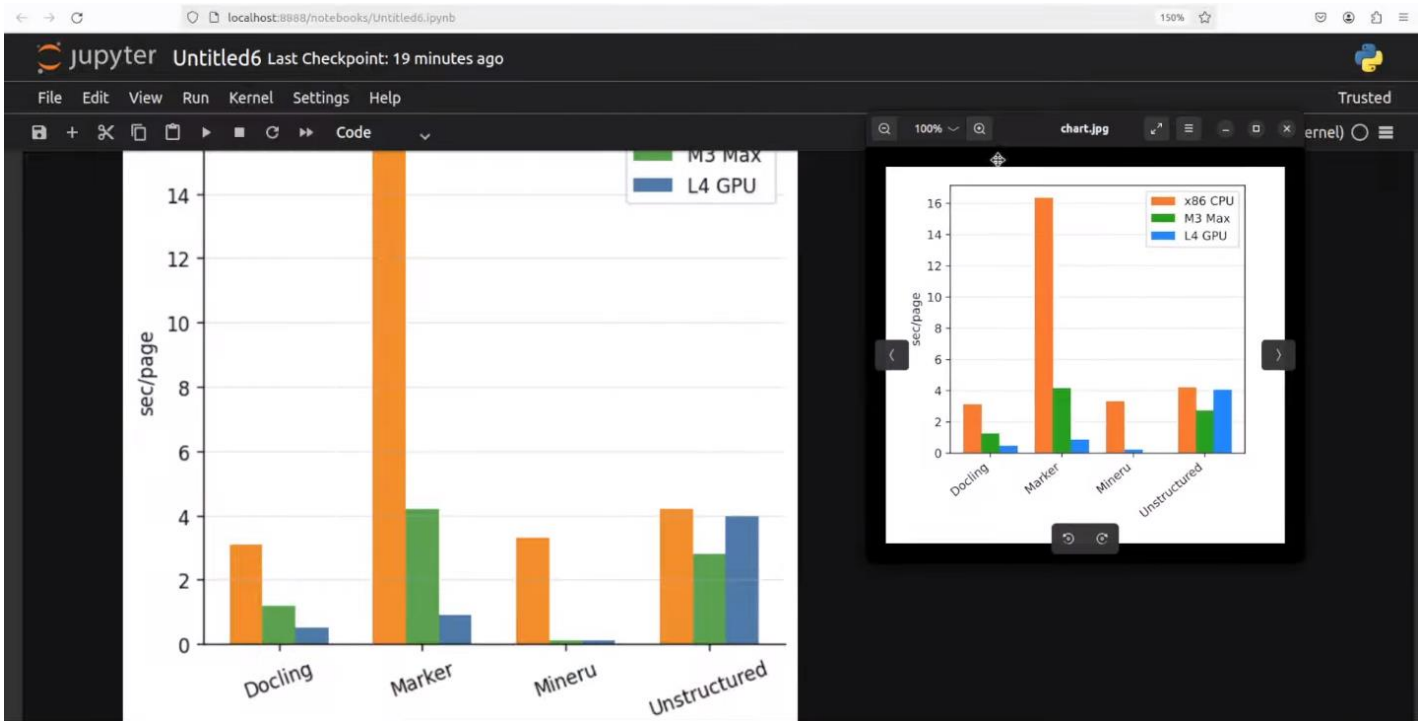
plt.figure(figsize=(5, 4.8), dpi=120)
plt.bar(x - w, x86, width=w, label="x86 CPU", color="#F28E2B")
plt.bar(x, m3, width=w, label="M3 Max", color="#59A14F")
plt.bar(x + w, l4, width=w, label="L4 GPU", color="#4E79A7")
```

```
plt.figure(figsize=(5, 4.8), dpi=120)
plt.bar(x - w, x86, width=w, label="x86 CPU", color="#F28E28")
plt.bar(x, m3, width=w, label="M3 Max", color="#59A14F")
plt.bar(x + w, l4, width=w, label="L4 GPU", color="#4E79A7")

plt.ylabel("sec/page")
plt.xticks(x, cats, rotation=20)
plt.ylim(0, 17)
plt.legend(loc="upper right", frameon=True)

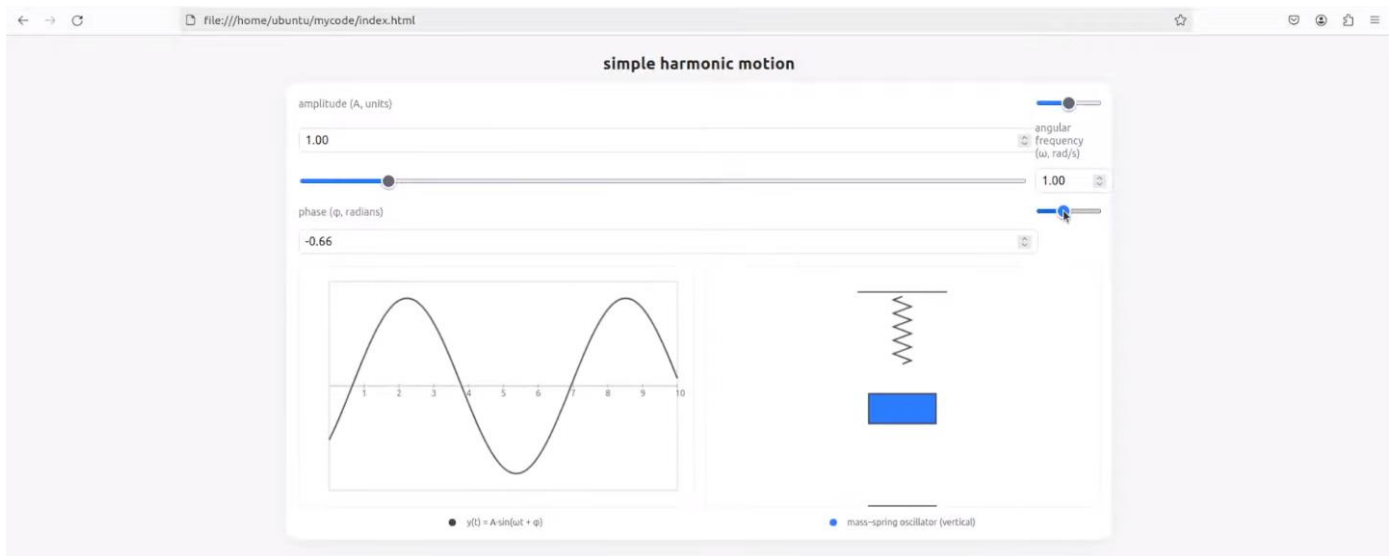
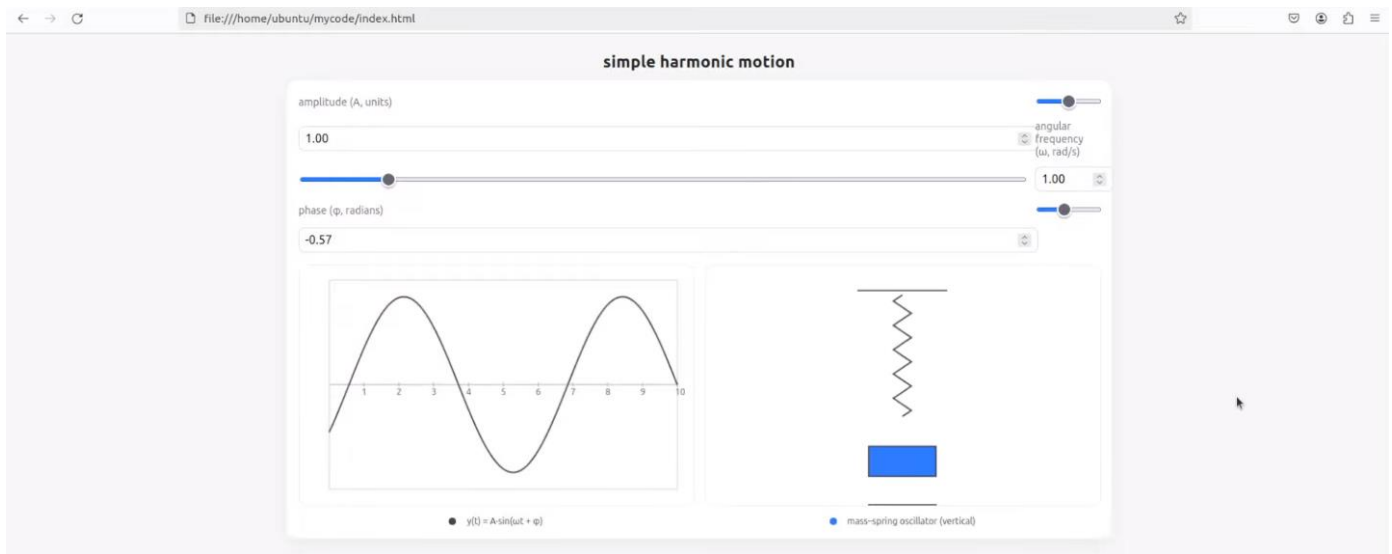
plt.grid(axis="y", linestyle="-.", linewidth=0.6, alpha=0.4)
for spine in ["top", "right"]:
    plt.gca().spines[spine].set_visible(False)

plt.tight_layout()
plt.show()
```



It was correct





It worked correctly as a single HTML file from the JanusCoderV-8B model.